Ariadne

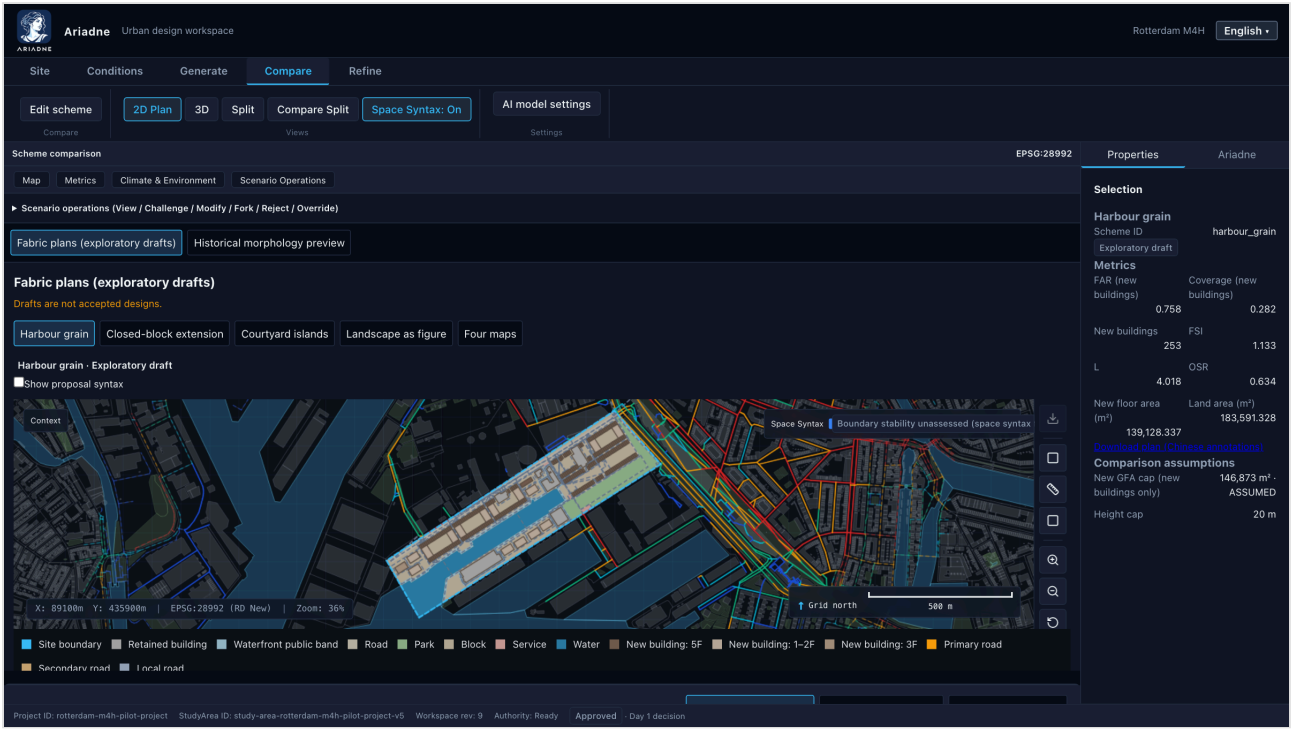
An Evidence-Bound, Contestable AI Partner for Early-Stage Urban Design

Course: IDEM307 Generative AI and Design, TU Delft (2026–27)

Group member: Yu Cheng (Jack) Hsieh (sole member)

Code: https://github.com/hycsposad113/ariadne

Statement: The course team may show the demo video publicly.



1. Motivation and objectives

Early-stage urban design fixes streets, blocks and building volumes, and with them daylight, microclimate and movement, for decades. Two gaps make AI hard to use at this stage. Chat models answer fluently and confidently, but without sources, and they invent numbers. Environmental simulations and street-network analyses are slow to run and hard to question, so their results arrive late and are taken on trust.

Ariadne is for an urban designer (student or practitioner) in the concept stage who wants an AI partner that reasons from evidence, states its limits and can be challenged, rather than an autopilot.

The tool is demonstrated on the Rotterdam Merwe-Vierhavens (M4H) harbour (183,591 m² of land). My own earlier studio design for the site ("Metabolic Engine": floor area ratio (FAR, built floor area divided by land area) ≈ 0.8, 4–6 storeys, about 40 % open space) is the human reference; its intent defines the constraints the generated schemes are checked against.

The opportunity is a partner that shows how it reached each judgement, so the designer can check it and push back, and slow analyses become something to argue with during the concept stage. The project's rule is: "*LLMs reason. Deterministic tools calculate. Validators verify. Humans decide.*"

2. Tool design and AI integration

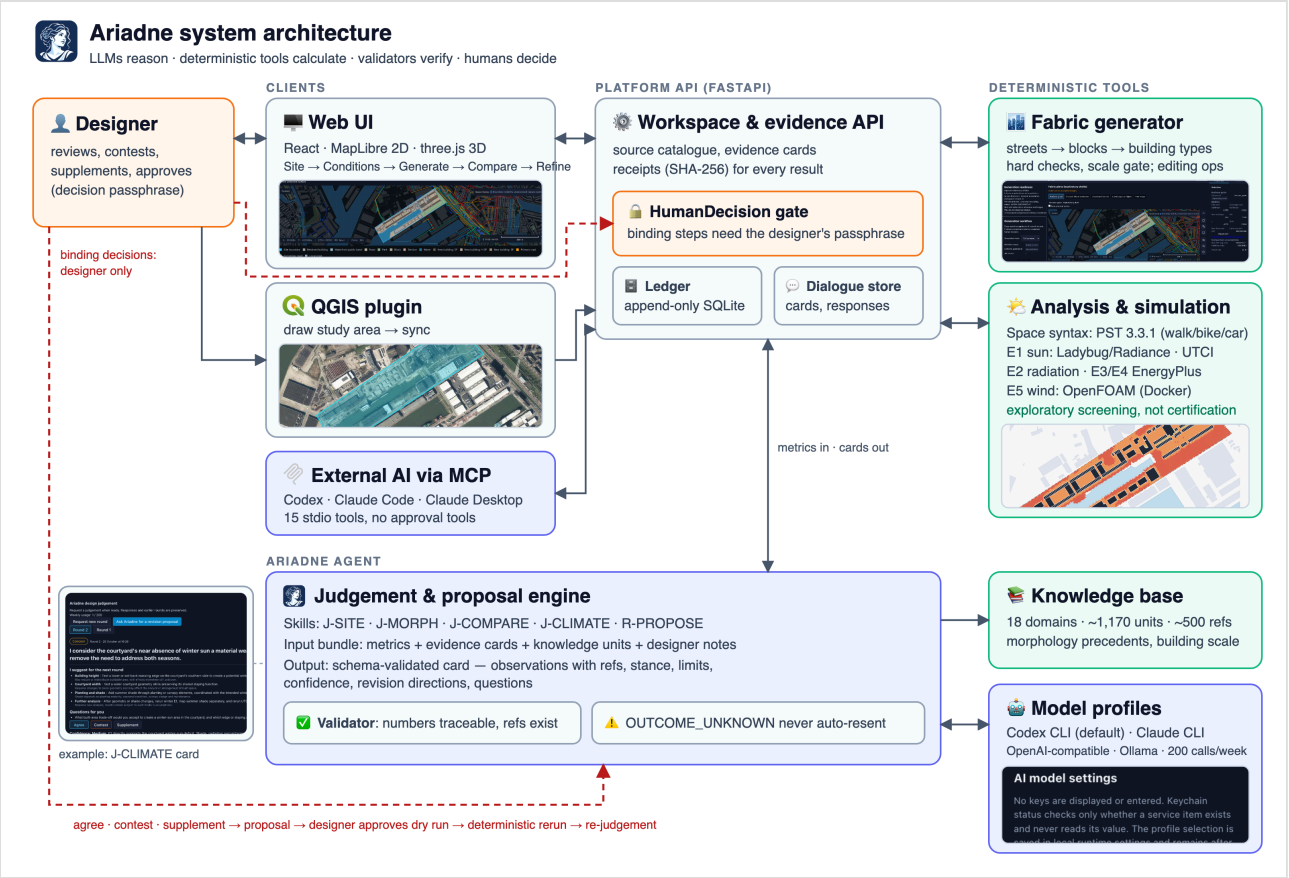


Figure 1: System architecture with screenshots of each part. The AI (blue) judges and proposes; deterministic tools (green) calculate; only the designer (orange) passes the HumanDecision gate. MCP (Model Context Protocol) is an open standard that lets an external AI assistant call a tool's functions.

Workflow

A React web interface talks to a FastAPI backend with an append-only ledger; a QGIS plugin draws the study area (Figure 1). Five tabs organise the work:

- **Site:** study area, buildings, street network and site analysis, including unknowns (Figure 2).
- **Conditions:** retained buildings, constraints, and space-syntax context. Space syntax measures how well connected each street is within the whole network, a proxy for how likely people are to pass through it.
- **Generate:** a deterministic generator builds four urban fabrics (*Harbour grain*, *Closed-block extension*, *Courtyard islands*, *Landscape as figure*): development axes shape the streets, blocks follow, then building types, with hard checks.
- **Compare:** 2D/3D, density metrics, space syntax and climate results (Figure 4).
- **Refine:** the designer edits buildings (floors, footprints, setbacks) and roads by hand; edits beyond the constraints are blocked.

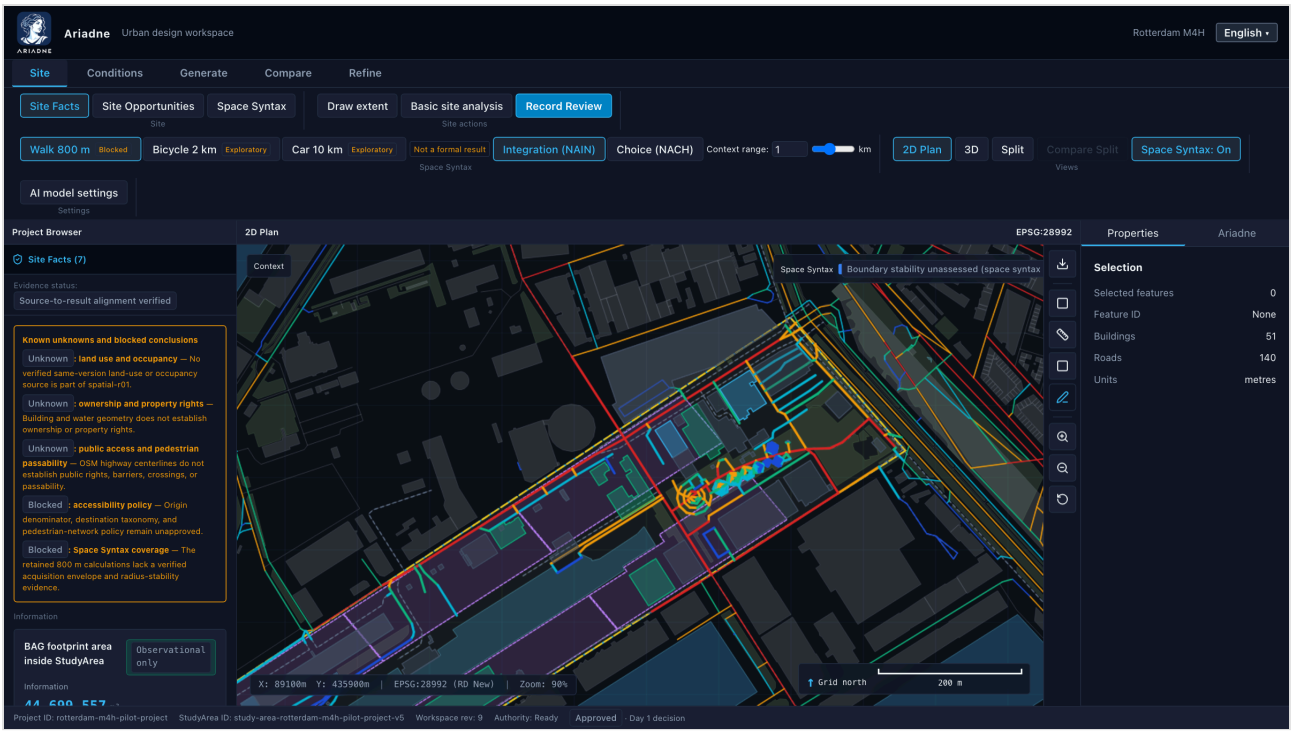


Figure 2: Site tab — M4H study area with buildings, street network and site analysis.

Deterministic environmental screening

All numbers come from established simulation tools, used as screening, not certification (Figures 5–10). All five methods were run for the four schemes and are shown in the interface with SHA-256 receipts; E1 and E2 can be re-run from the interface, the others run as offline scripts.

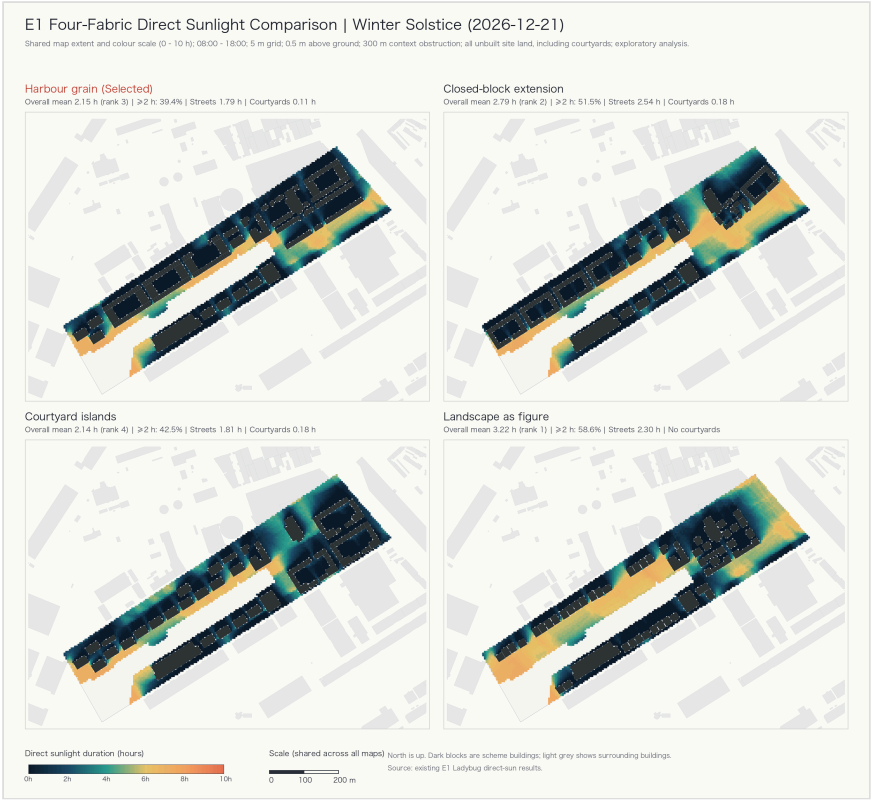


Figure 5: E1 — hours of direct sunlight on the ground on the winter solstice for the four fabrics (Ladybug/Radiance). Shows which open spaces get winter sun.

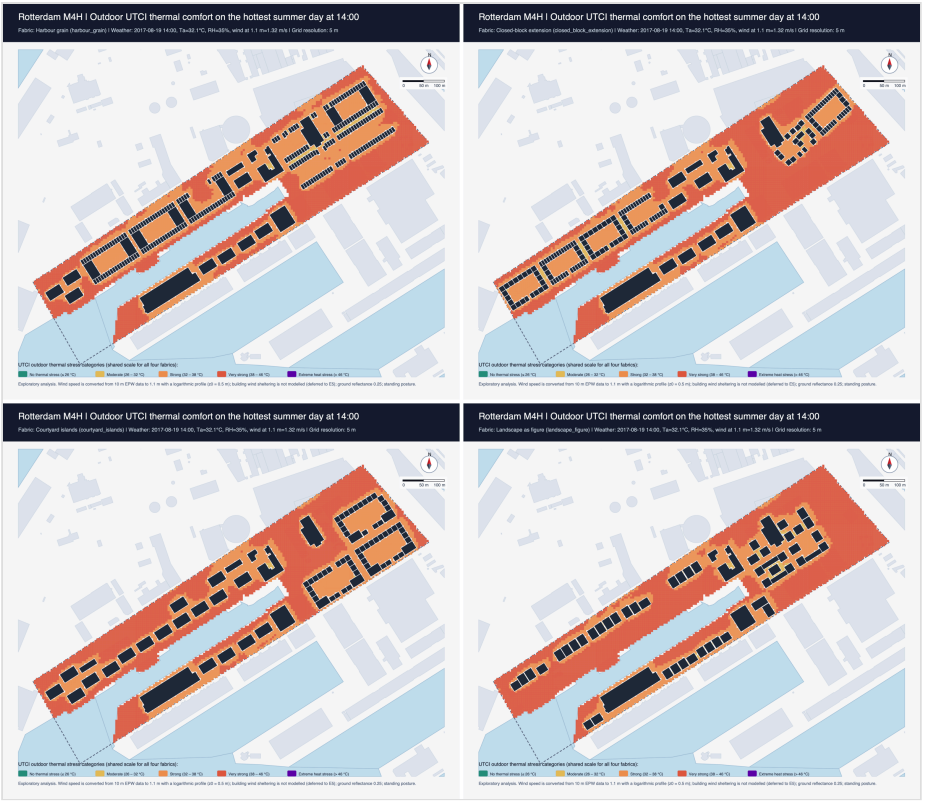


Figure 6: UTCI (Universal Thermal Climate Index) — how hot it feels outdoors at 14:00 on the hottest summer day, combining air temperature, radiation, wind and humidity, for the four fabrics.

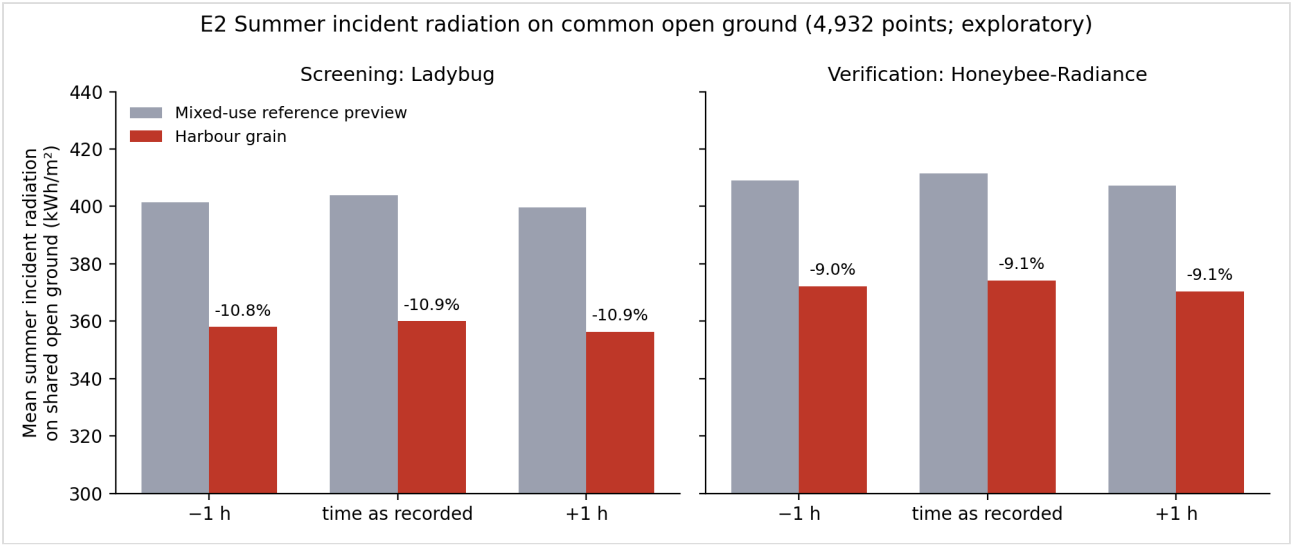


Figure 7: E2 — summer solar radiation reaching shared open ground, Harbour grain against a mixed-use reference scheme, checked with two methods and ±1 h; Harbour grain's buildings shade the ground about 9–11 % more. Axis starts at 300 kWh/m².

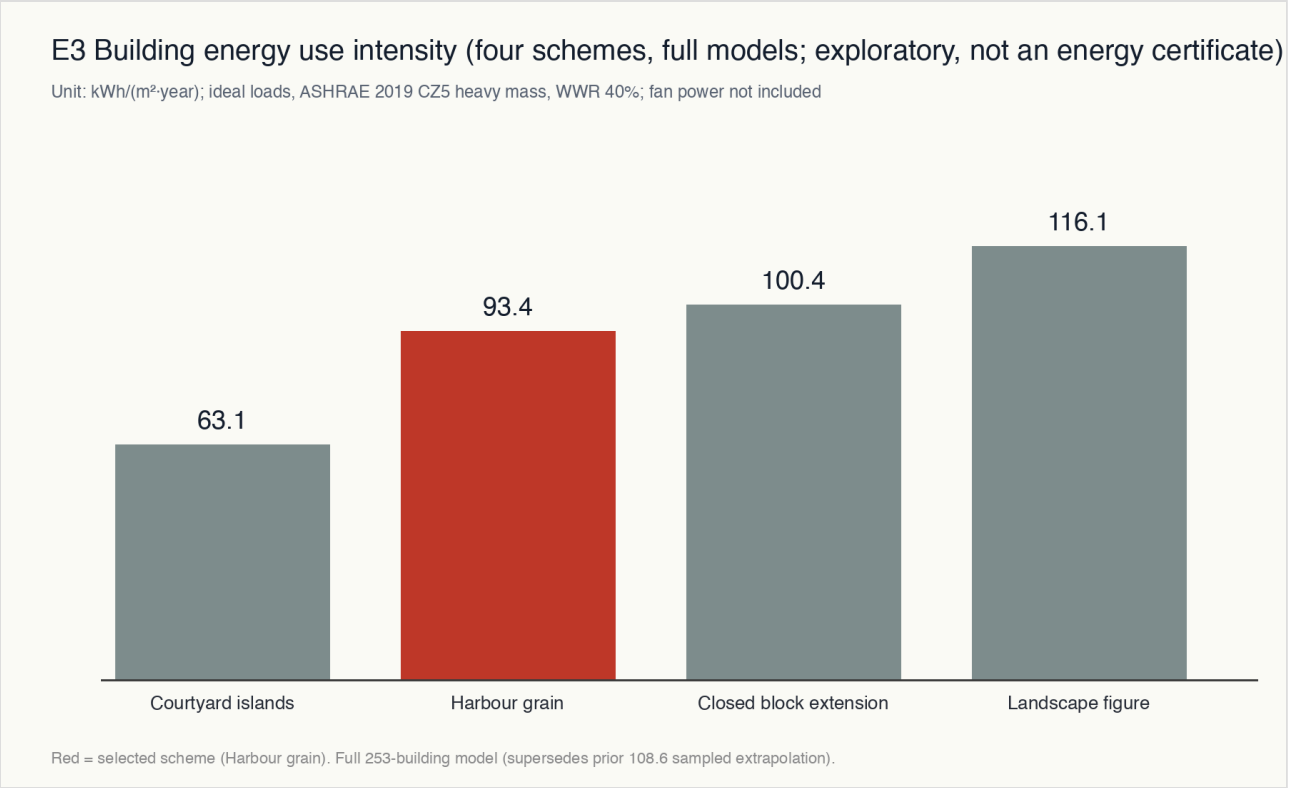


Figure 8: E3 — building energy use intensity (EUI: yearly heating and cooling energy per m² of floor area), EnergyPlus ideal-load models of all buildings in each scheme.

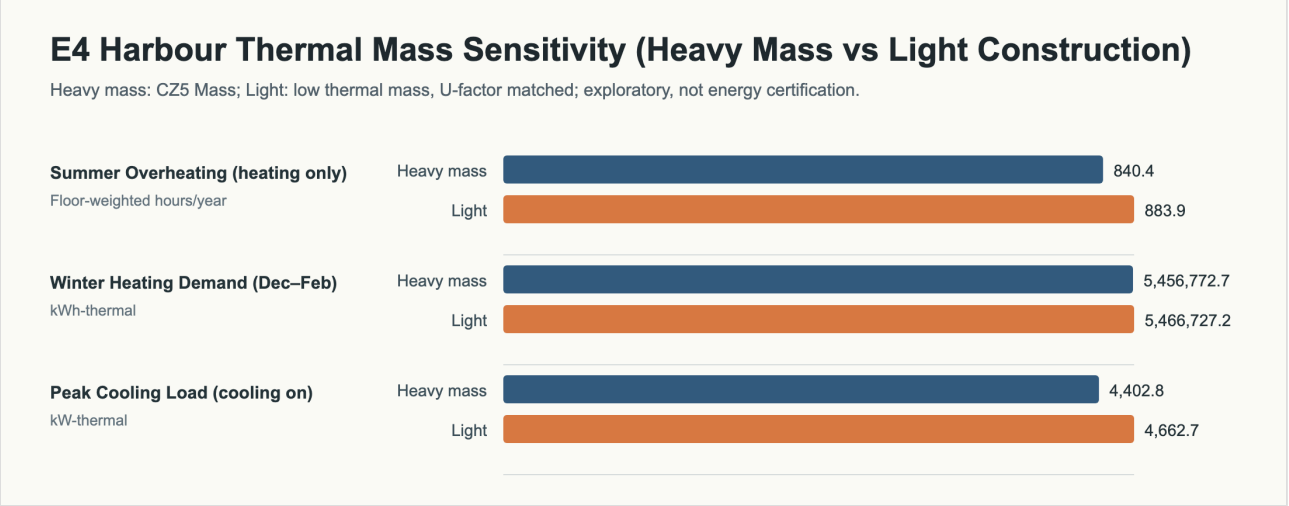


Figure 9: E4 — thermal mass: heavy versus light construction for Harbour grain, i.e. how much a heavy structure buffers summer overheating and winter heating demand.

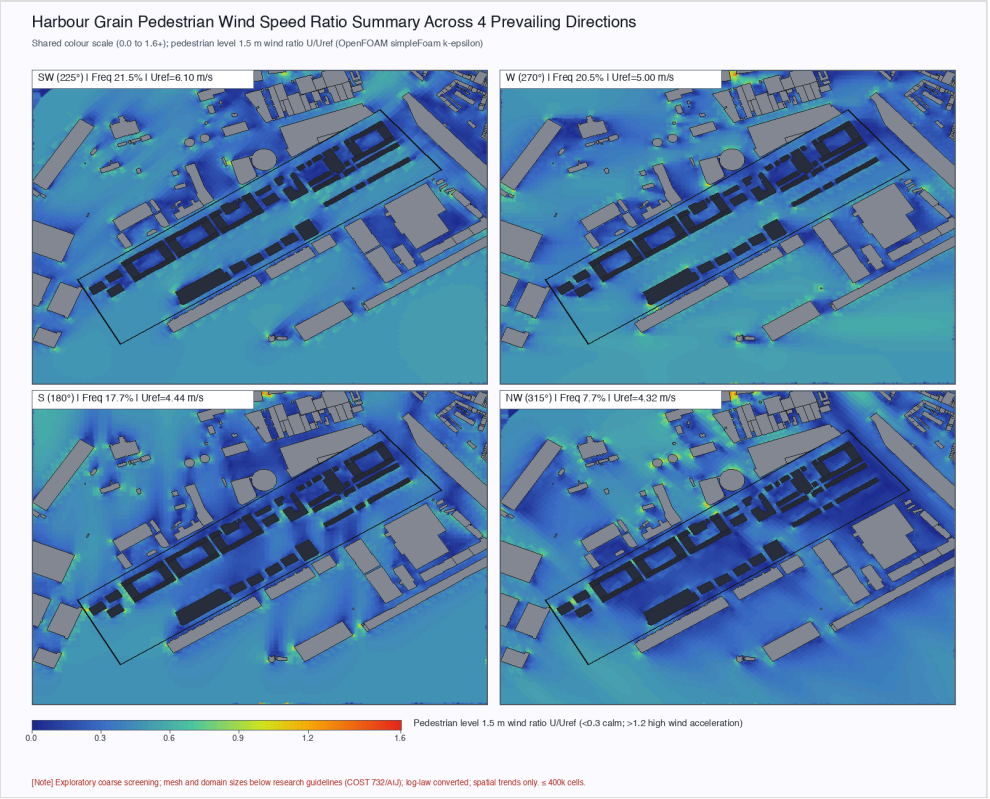


Figure 10: E5 — pedestrian-level wind speed relative to the reference wind for Harbour grain under the four prevailing winds (coarse OpenFOAM computational fluid dynamics); blue = sheltered, yellow–red = accelerated.

Ariadne's skills

Ariadne works only through skills: fixed instructions that define the inputs (computed metrics, evidence cards, knowledge units, designer notes), the judgement framework and the citation rules.

Skill	What it does
J-SITE	judges the site analysis: what is known, unknown, and decided
J-MORPH	judges a scheme's urban form: block size, courtyards, street walls, scale
J-COMPARE	compares schemes and names the trade-offs
J-CLIMATE	interprets the simulations for a scheme
R-PROPOSE	drafts a revision proposal after the designer contests a card

The output is a schema-validated "judgement card" (Figure 3): observations with metric references, a stance, reasoning, counterpoints, limitations with named data gaps, confidence and questions for the designer. A validator rejects numbers that cannot be traced to the inputs; a call whose delivery is unknown is recorded and never resent automatically. A knowledge base of 18 domains (about 1,170 units, 500 verified references) supplies the design knowledge, including building-scale common sense.

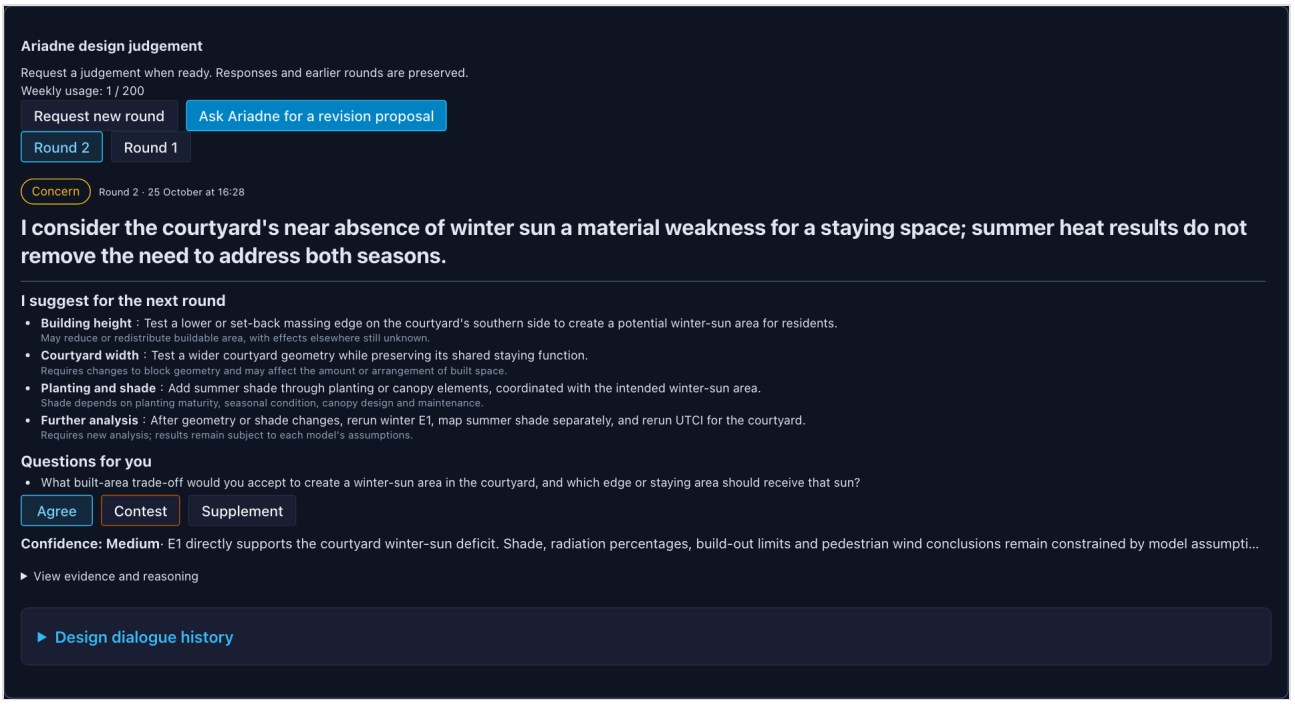


Figure 3: A J-CLIMATE judgement card: stance, headline, suggestions for the next round, a question for the designer, and Agree / Contest / Supplement.

Dialogue and the M4H example

The designer answers a card with Agree, Contest or Supplement. On request, Ariadne drafts a proposal from fixed operations (e.g. a solar envelope that lowers south-facing wings), describing effects only qualitatively. If the designer approves the dry run, the platform recalculates and Ariadne re-judges against the before/after numbers. Proposals never overwrite the accepted scheme.

In the M4H run, J-CLIMATE judged the near-absence of winter sun in the Harbour-grain courtyards a material weakness (mean 0.108 h of direct sun; 1.03 % of the courtyard area reaches 2 h) and asked which trade-off I would accept. I had not checked this myself. I answered that the courtyards are residents' staying spaces and summer shade, and that winter sun and summer comfort matter equally; that answer became a design condition stored for later judgements. In the run recorded for the demo, a solar-envelope proposal lowered the south wings by up to two floors. The courtyard mean rose from 0.11 h to 0.22 h while site-wide very strong heat stress rose from 26.4 % to 26.7 %; Ariadne called this a "partial response". I agreed: the direction is right, but 0.22 h is far from a usable winter sitting area. I did not accept the draft and asked where widening a courtyard could create a sunny edge without breaking the harbour street wall.

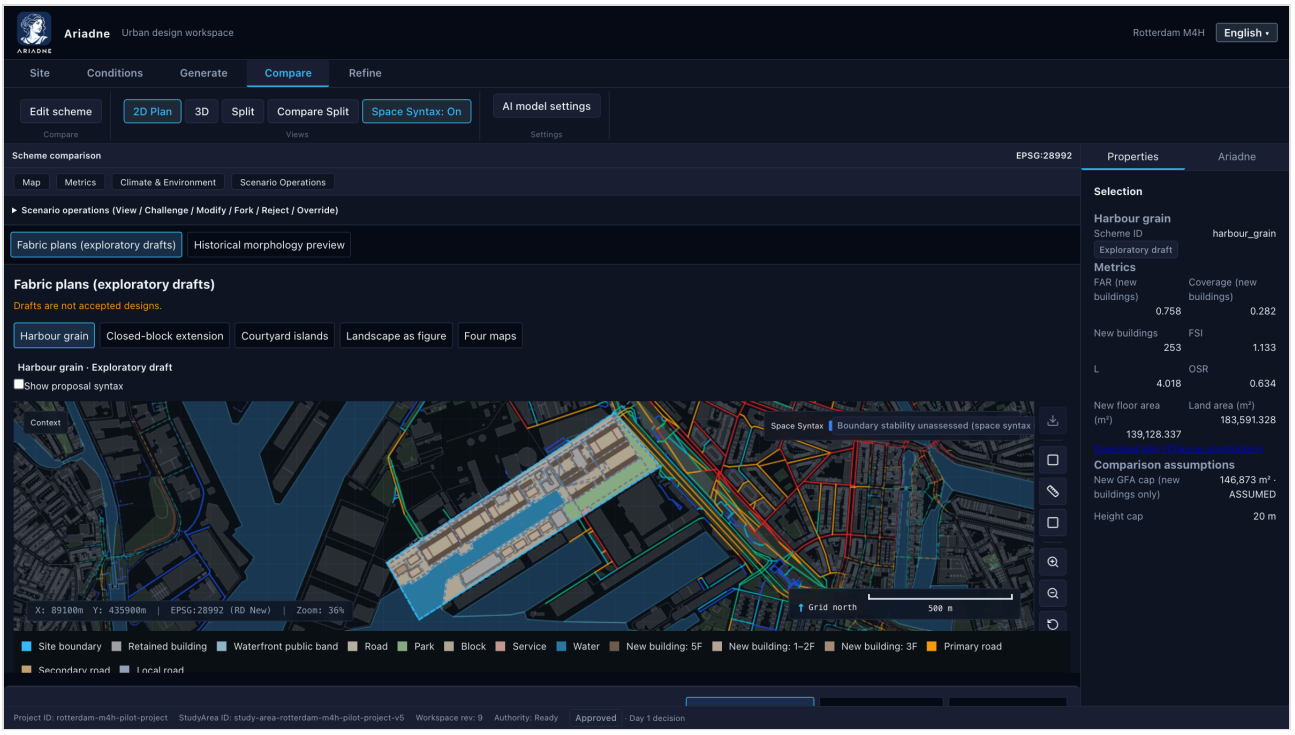


Figure 4: Compare tab — Harbour grain (exploratory draft) with the space-syntax overlay (warmer colours = better-connected streets) and its metrics.

Models and bring-your-own AI

The designer chooses Ariadne's model in an "AI model settings" panel: Codex CLI (default), Claude Code CLI, or an OpenAI-compatible API whose key stays in the macOS Keychain. The panel shows whether a profile is usable; live calls are capped at 200 per week. Through MCP, an external assistant (Codex, Claude Code, Claude Desktop) gets 15 tools in English or Chinese, without any approval tool. Binding steps (study area, retained buildings, constraints, accepting a scheme) need the designer's local passphrase.

Implemented vs conceptual

Area	Implemented	Partly implemented / conceptual
Site	M4H data package; QGIS study-area plugin.	Drawing any site and fetching its data automatically.
Generation and editing	Four generated fabrics; manual editing of buildings and roads.	AI proposals that change the street network (they change buildings only).
Simulations	All five methods run, results shown in the interface; E1/E2 re-run after an edit.	Re-running UTCI and E3–E5 from the interface after an edit; research-grade wind modelling.
Ariadne	5 skills; validator; proposals; dialogue; model choice; 15 MCP tools.	A verified Claude-driven run (CLI not logged in on the demo machine); multi-user use.

3. Development process

I built Ariadne between 25 September and 26 October 2026 (about 500 commits), writing little of the code myself. I acted as designer and decision-maker; a frontier model (Claude Opus) acted as

commander and reviewer; cheaper models (Codex, Gemini, Grok, Claude Haiku) wrote code in isolated branches. Rules: code from one vendor's model is reviewed by another's; agents never read credentials; nothing is deleted; I decide at fixed review points. About 3,800 automated tests and browser walkthroughs in both languages back this up.

Decisions driven by my critique

1. **Scale before metrics:** the first massing looked like small warehouses, not housing: 7–8 m deep wings, 28 × 28 m courtyards, uniform heights. The AI had optimised network checks and never looked at building scale. A scale check (wing depth by use, courtyards ≥ 20 m, height variation) now runs before any plan is shown.
2. **Blocks before roads:** three rounds of "roads first, then blocks" produced almost no continuous street walls and fragmented parks. I inverted the order: development axis, 60–100 m perimeter blocks, block edges becoming streets, remaining land becoming named parks.
3. **Knowledge before generation:** rule-based generation was monotonous; AI-written design intents put landmarks in the same places; a later round looked random, with sharp corners and land left without a use. I had the knowledge base rebuilt and fixed a sequence: read the context, choose one morphological concept per scheme from measured precedents, then streets, blocks and building types.
4. **Testing the knowledge:** in one blind-marked set of graduate-level questions, the model scored 15.57/20 with the knowledge base and 14.15 without, mainly through better evidence and honesty. Limited evidence, but it showed where the base had gaps.
5. **Judgement as cards:** the AI's critique varied with the model, so judgement moved into fixed skills and validated cards.

AI shortcomings and responses

- **Overstated completion:** agents reported a release as finished after testing only Chinese and 2D; in English the interface showed Chinese and the 3D button did nothing. Every release now checks both languages, every view and the console.
- **Rule-breaking:** despite the no-deletion rule, agents silently deleted their own folders; every task prompt now lists prohibitions and requires disclosure.
- **Output drift:** English cards came back in Chinese and numbers appeared without sources; language-aware prompts and the validator followed.
- **Fragile evidence:** space-syntax records were stored in a temporary folder the system cleared, so the comparison layers failed. Recalculating reproduced all 12 results exactly; a local wind rerun also matched exactly.
- **Reviews catch AI errors:** a second model's review of documentation written by another found 27 false or overstated claims; a review of this report found placeholder numbers in the MCP tools, since removed.
- **Shared quota:** Ariadne and the coding agents share one model account; when the agents used up the quota, Ariadne stopped working too.

4. Critical reflection

The tool

Compared with my studio process, the biggest difference is speed and attention. Four schemes and their simulation numbers appear at once, and Ariadne points at problems I had not checked. The courtyard card changed my design: it made me state that courtyards must have winter sun and summer shade, and that statement became a condition the next proposals had to meet. The cost is time spent judging whether the AI is wrong.

The limits are real. The release works on one site, simulations are screening and partly offline, and Ariadne cannot judge what is not measured: tree shade, for example, appears only as a data gap. For practice I would want any site with automatic data, simulations that update after every edit, AI proposals that can change the street network, several people working together, and one more step: turning a designer's rough sketch layout directly into a plan that is simulated and analysed within minutes, so the sketch itself can be adjusted.

AI-assisted development

AI let me, a designer rather than a programmer, build a working, tested platform in a month. But its knowledge of the physical world needed constant correction. The "warehouse" wings passed every network check; only my first look at the plan caught them, and the fix was to turn that look into a rule the machine checks. Agents also overstated their own progress, so much of the month went into verification rather than design, and that stretched the schedule.

Designers building their own AI tools

A designer's leverage lies less in writing code than in deciding what the AI may judge, which evidence it must cite, what it may never do, and when the designer looks. The hardest part was translating tacit knowledge (scale, enclosure, morphology) into rules and knowledge the AI can use. A designer can build a serious AI tool this way, but only with strict verification habits and by keeping the decisions.

References

- Hillier, B. and Hanson, J. (1984) *The Social Logic of Space*. Cambridge: Cambridge University Press.
- Spatial Morphology Group, Chalmers University of Technology (n.d.) *Place Syntax Tool*, version 3.3.1. Available at: https://github.com/SMoG-Chalmers/PST.
- Ladybug Tools (n.d.) *Ladybug Tools*, with Radiance 6.0.2. Available at: https://www.ladybug.tools/.
- Bröde, P. et al. (2012) 'Deriving the operational procedure for the Universal Thermal Climate Index (UTCI)', *International Journal of Biometeorology*, 56, pp. 481–494.
- OpenCFD Ltd (2025) *OpenFOAM*, version v2506. Available at: https://www.openfoam.com/.
- U.S. Department of Energy (n.d.) *EnergyPlus*. Available at: https://energyplus.net/.
- Anthropic (2024) *Model Context Protocol specification*. Available at: https://modelcontextprotocol.io/.

- TU Delft (2026) *IDEM307 Generative AI and Design: Group project — final submission and assessment*. Delft: Delft University of Technology.
- Hsieh, Y. C. (2026) *Metabolic Engine*. Urbanism Q2 studio portfolio, TU Delft (unpublished).

Word count (Sections 1–4, excluding the title page, figure captions and references): 1,745.